

Sample 67955: A description and a problem

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Abstract—Clasts in breccia sample 67955 contain relatively coarse olivine and plagioclase in part enveloped by poikilitic hypersthene and diopside. This distinctive texture is concluded to result from high temperature crystallization in the presence of a fluid phase, probably a silicate melt. A problem posed by this rock is whether the bulk chemistry, characterized by a high Mg/Fe ratio but low total Fe+Mg, is the result of primary igneous processes on the moon or whether it is the residual end product of a multiple distillation process resulting from repeated meteorite impacts.

INTRODUCTION

THIS NOTE describes clasts in a breccia sample, 67955,22, most of which exhibit a similar, but coarser, poikilitic texture to that described by Albee *et al.* (1973), Bence *et al.* (1973), and Simonds *et al.* (1973). One important question posed by the poikilitic rocks is whether their mineralogy and texture are the result of high temperature metamorphic (Albee *et al.*, 1973; Bence *et al.*, 1973) or igneous (Simonds *et al.*, 1973) processes. The distinction is important because if the poikilitic rocks can be shown to be the result of crystallization from a silicate melt near liquidus temperature they may give direct information on lunar crustal processes as distinct from lunar surface (e.g., impact) processes.

DESCRIPTION OF SAMPLE 67955

Texture

The overall general appearance of rock 67955 is very similar to that described by Short (1966) for shock-lithified breccias found at nuclear bomb crater sites. The clasts, up to 1.2 mm, are typically sub-rounded and are set in a glassy matrix containing angular lithic and mineral fragments that, based on chemical similarities of similar minerals, appear to be related in provenance to the larger clasts. The clasts themselves are difficult to resolve from the matrix because of their similar mineralogical content and because the abundant feldspar has been so strongly affected by shock as to destroy any original textural differences between the feldspar of the clasts and that of the matrix.

The bulk composition is gabbroic anorthosite (a bulk analysis is given by LSPET, 1972); 78.5 modal percent of the rock is (or was) plagioclase feldspar; the remaining essential minerals are pyroxene (14.5%, hypersthene more abundant

than diopside), olivine (6%), and opaques (1%) which are predominantly ilmenite and metal alloys. The high Mg/Fe ratio of the rock is reflected in the high Mg/Fe ratio of the mafic silicates (Table 1).

The last brecciation and shock apparently did not affect the mafic minerals in the clasts or the smaller fragments in the matrix. (Representative analyses of minerals in the clasts and individual mineral fragments are given in Table 1.) For example, the brecciation event produced small, angular rock fragments, now occurring in the matrix, which clearly result from fracture through and fragmentation of mafic mineral associations similar to those in the clasts. The compositions of the minerals in the small fractured fragments are essentially the same as those in minerals with identical texture in the larger clasts (compare analyses of fragment G, Table 1, with those of clast A, Table 1). Further, there is no chemical gradient in the olivine, pyroxene, or plagioclase of the broken fragments (for example, fragment G) towards the contact of the crystals and the matrix, as would be expected if post-fracturing thermal metamorphism resulting from the last shock had any effect on the mineral chemistries.

The typical mineral assemblage in the clasts consists of a single poikilitic (or poikiloblastic) pyroxene enclosing discrete grains of olivine (0.03 to 0.06 mm) and plagioclase (0.02 to 0.06 mm). The poikilitic pyroxene, which occurs as either calcic or subcalcic varieties, may extend across an entire clast (up to 1.2 mm) or may occupy only a small part of a clast, the rest being plagioclase. The arms of the poikilitic pyroxene are up to 0.05 mm thick and thus are markedly thicker than those described by Bence *et al.* (1973) for other Apollo 16 poikilitic pyroxene-bearing rocks. In addition to the olivine enclosed in the pyroxene, spherical to sub-spherical olivine grains less than 0.03 mm in diameter occur in the plagioclase without associated pyroxene. A few clasts also contain up to a few percent of ilmenite and metal alloys.

Because the clasts have variable relative proportions of the several mineral components (for example, plagioclase ranges between 60 and 100%), they could be considered as coming from different source rocks. However, the minerals common to the different clasts have essentially the same chemistry (Table 1), and the total number of different minerals in the rock taken as a whole does not exceed the apparent number of chemical components. Therefore, the clasts are probably genetically related to each other and probably also came from the same source area.

Plagioclase

The plagioclase in the clasts shows a range of shock effects that vary from none visible to complete vitrification. It has a composition varying from An_{92} to An_{95} . This compositional range is found in each clast; it does not vary outside these limits between clasts. No apparent relation of these chemical variations to shape of plagioclase grain (i.e., normal zoning) or proximity to a mafic mineral was detected.

Table 1. Selected microprobe analyses of phases in clasts and mineral fragments in breccia 67955.

Clast A				Clast C				Clast F			Fragment G			Mineral Fragments		
Mineral	Hyp.	Ol. (1)	Ol. (2)	Plag.	Aug.	Ol. (1)	Plag. (1)	Plag.	Ol. (2)	Plag.	Hyp.	Ol. (1)	Plag.	Il.	Ol. (3)	Plag.
SiO ₂	55.1	38.6	38.4	44.3	50.9	38.6	44.9	45.7	38.7	44.5	55.3	39.1	45.1	0.1	39.3	44.7
TiO ₂	0.73	0.07	0.04	0.05	2.19	0.06	0.06	0.05	0.03	0.03	0.79	0.06	0.07	52.2	0.04	0.05
Al ₂ O ₃	0.76	—	—	35.3	2.43	—	35.2	35.6	0.01	34.9	0.88	—	35.1	—	—	34.1
Cr ₂ O ₃	0.43	0.04	0.04	—	0.88	0.05	—	—	0.05	—	0.45	0.04	—	0.2	0.04	—
FeO	12.2	20.8	20.6	0.10	6.19	21.6	0.14	0.05	20.2	0.09	12.1	20.3	0.15	39.4	18.4	0.06
MnO	0.22	0.21	0.21	—	0.17	0.26	—	—	0.23	—	0.21	0.22	0.01	0.5	0.06	0.01
MgO	28.6	41.1	40.7	0.05	16.5	39.4	0.01	0.02	40.7	0.04	27.1	39.8	0.02	5.5	42.2	0.04
CaO	1.78	0.05	0.14	19.5	20.7	0.07	19.4	19.4	0.11	19.3	1.77	0.06	18.8	0.2	0.06	18.4
Na ₂ O	—	—	—	0.54	0.09	—	0.65	0.84	—	0.55	—	—	0.71	—	—	0.70
K ₂ O	—	—	—	0.05	—	—	0.07	0.09	—	0.07	—	—	0.07	—	—	0.09
Total	99.8	100.9	100.1	99.9	100.1	100.0	100.4	101.8	100.0	99.5	98.6	99.6	100.0	98.1	100.1	98.2

Cation Proportions

Si	1.97	0.99	0.99	2.05	1.88	1.00	2.07	2.08	0.99	2.07	1.97	1.01	2.08	—	1.00	2.10
Al	0.03	—	—	1.93	0.11	—	1.91	1.90	—	1.91	0.04	—	1.91	—	—	1.89
Ti	0.02	—	—	—	0.06	—	—	—	—	—	0.02	—	—	0.97	—	—
Cr	0.01	—	—	—	0.03	—	—	—	—	—	0.01	—	—	—	—	—
Fe	0.37	0.45	0.44	—	0.19	0.47	—	—	0.43	—	0.37	0.44	0.01	0.82	0.39	—
Mn	0.01	—	—	—	0.01	—	—	—	—	—	0.01	0.01	—	0.01	—	—
Mg	1.54	1.57	1.57	—	0.91	1.52	—	—	1.57	—	1.51	1.54	—	0.20	1.60	—
Ca	0.07	—	—	0.97	0.82	—	0.96	0.95	—	0.96	0.07	—	0.93	0.01	—	0.93
Na	—	—	—	0.05	0.01	—	0.06	0.07	—	0.05	—	—	0.06	—	—	0.06
K	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Notes: Mineral abbreviations: Hyp. = hypersthene; Ol. = olivine; Plag. = plagioclase; Aug. = augite; Il. = ilmenite.
(1) denotes mineral enclosed by pyroxene.
(2) denotes olivine spherule in plagioclase.
(3) large olivine fragment.

Olivine

In the clasts, the olivine occurs as tiny (up to 0.03 mm), remarkably perfect little spheres or as larger (over 0.03 mm and up to 0.06 mm) subhedral, equant grains. The larger grains are typically enveloped by the poikilitic pyroxene; the smaller spheres are not associated with pyroxene but occur alone as inclusions in plagioclase. The olivine is more homogeneous than the plagioclase: olivine surrounded by orthopyroxene or plagioclase, in most clasts, has the same composition, Fo₇₈ (Table 1). The olivine surrounded by clinopyroxene is slightly more iron-rich, Fo₇₇. (Another exception, a single, relatively large (0.5 mm) olivine fragment, has the average composition Fo₈₀. This clast, and another exceptional clast of Fe–Ni, are described below). All of the olivine is very low in the minor elements (Cr as Cr₂O₃ ≈ 0.04%, CaO ≈ 0.06%, Ti as TiO₂ ≈ 0.04%), suggesting they did not grow directly under surface lava conditions (Smith, 1971).

Pyroxene

The pyroxene in the clasts has a poikilitic habit as described for other Apollo 16 rocks (LSPET, 1972). In contrast with those poikilitic pyroxenes described by Bence *et al.* (1973), the individual arms of the poikocrysts are relatively thick, over 0.05 mm, and the equant grains of plagioclase or olivine enclosed within the arms are up to 0.06 mm in diameter. Because the olivine crystals have about the same composition whether they are enclosed in pyroxene or are isolated in plagioclase, the pyroxene is probably not a product of chemical reaction between olivine and plagioclase. This conclusion may also be drawn from the fact that the pyroxene may be either clinopyroxene or orthopyroxene.

Discrete grains of both calcic and subcalcic pyroxenes were not found in the same clast. However, thin, parallel lamellae of sub-calcic pyroxene were identified by X-ray area scan techniques in one calcic pyroxene grain, suggesting both low Ca and high Ca pyroxene phases may coexist; the lack of such association in the thin section may therefore be fortuitous. Furthermore, the nearly uniform compositions of plagioclase, olivine, clinopyroxene, and orthopyroxene in all clasts suggest all four minerals may have originally been in mutual chemical equilibrium.

The pyroxenes are conspicuously homogeneous considering the extreme zoning observed in the pyroxenes of the mare lavas. They are as homogeneous as those of the “genesis” rock, 15415, and some of the pyroxene clasts in the breccias from other Apollo localities. The highly magnesian hypersthene composition En₇₉Wo₃Fs₁₈, combined with its low Al content, is a characteristic of hypersthene found as fragments in breccias and soils from several Apollo localities (Fig. 1a). The diopside (this name for the calcic clinopyroxene is used to emphasize its high magnesium and calcium content) is unusually magnesian and calcic for lunar clinopyroxenes, but is not extreme amongst those already described for non-mare rock types (Dowty *et al.*, 1972).

As for many of the pyroxenes from the lunar rocks, the relative content of Ca,

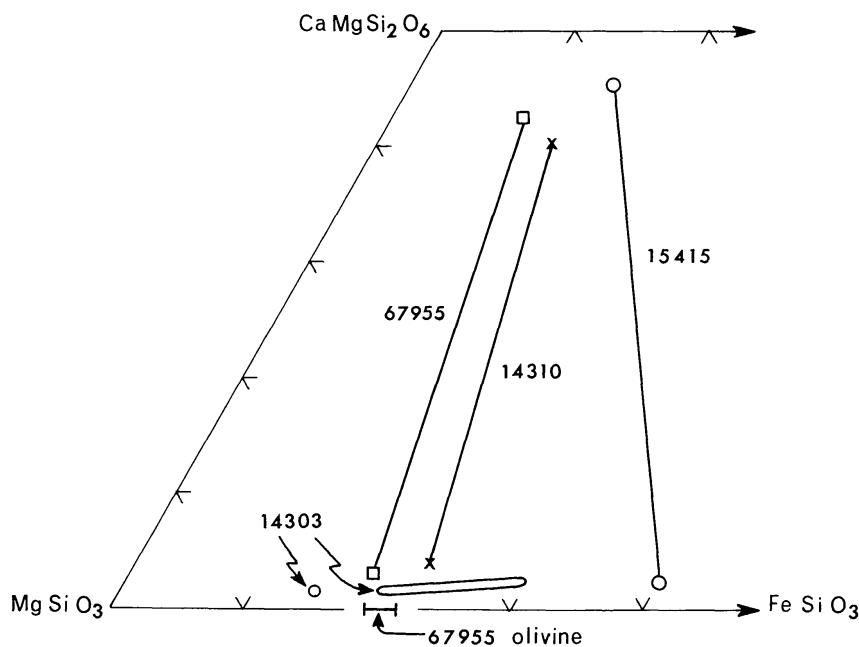


Fig. 1a. Portion of pyroxene quadrilateral diagram showing tielines between hypersthene-clinopyroxene pairs in samples 15415 (Hargraves and Hollister, 1972), 14310 (Hollister *et al.*, 1972), and 67955 (four clinopyroxenes and two orthopyroxenes plot within the appropriate boxes). Analyses of hypersthene fragments in breccia sample 14303 (Weigand and Hollister, 1972) are shown for reference. Also illustrated is the range of olivine compositions in sample 67955.

Fe, and Mg is not a sufficiently unique value for characterizing the provenance of the pyroxenes. It has already been shown (Hollister, 1972; Weigand and Hollister, 1972; Bence and Papike, 1972) that pyroxenes otherwise indistinguishable in provenance in terms of the major components Ca, Mg, and Fe are clearly distinguished in terms of the relative Cr, Ti, and Al contents. In Fig. 1b are plotted the relative Al, Ti, and Cr values for the pyroxenes from sample 69755 compared with those for several lunar sample groups. The data cluster in the group typical of pyroxenes in many breccia samples and soil fragments (Weigand and Hollister, 1972); this group is distinct from all basalts, including the hypersthene-bearing basalt 14310 which plots in the high Al region of the low Ca clinopyroxene field (Hollister, 1972).

The distribution function, K_D , of Fe and Mg between the diopside and the hypersthene, assuming they coexist, suggests a moderately high temperature of chemical equilibration of the two pyroxenes. The average distribution function, K_D , ($K_D \equiv (\text{Mg/Fe})_{\text{Hyp.}}/(\text{Mg/Fe})_{\text{Aug.}}$) for sample 67955 is 0.85. Typical values of K_D between the same minerals in terrestrial plutonic and extrusive rocks ($T \geq 900^\circ\text{C}$) generally exceed 0.70, and in terrestrial metamorphic rocks ($T \leq 800^\circ\text{C}$) they are generally less than 0.70 (Hermes, 1970). For reference, on the moon, K_D for the genesis rock (15415), described as a reequilibrated plutonic rock (Hargraves and Hollister, 1972) or as simply a metamorphic rock (James, 1972), is about 0.55, and that between hypersthene and augite in basalt 14310, possibly quenched near

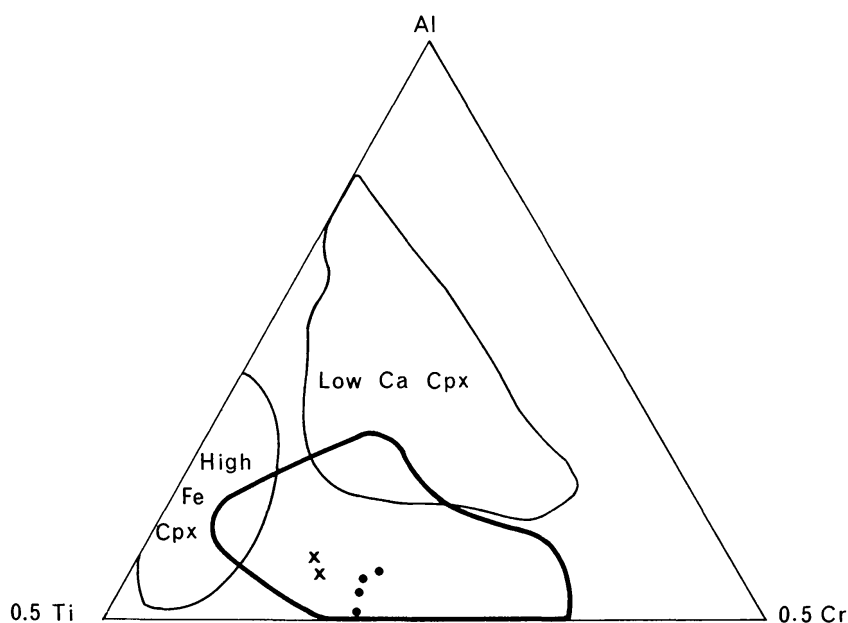


Fig. 1b. Al-Ti-Cr diagram (Hollister *et al.* 1971; Hollister, 1972) illustrating relative atomic proportions of these elements in lunar pyroxenes. Shown are the data for the clinopyroxene (x) and the orthopyroxene (.) in sample 67955. The heavy line outlines data of orthopyroxenes and high Ca clinopyroxenes in Apollo 11 and 12 soils, sample 15415, and two Apollo 14 breccias, 14063 and 14303. The two other fields shown for reference encompass analyses of clinopyroxenes from mare lavas. Data for orthopyroxene cores in 14310 (Hollister, 1972) plot within the low Ca Cpx field.

liquidus temperature, is 0.93 (data in Hollister, 1972; Hollister *et al.*, 1972). The width of the solvus in the pyroxene quadrilateral (Fig. 1a), defined by the Ca contents of the coexisting pyroxenes, also suggests the relative order for temperatures of equilibration of pyroxenes as 14310 > 67955 > 15415.

Mineral fragment clasts and metals

In addition to the pyroxene-olivine-feldspar clasts, there are also the single olivine clast mentioned above and a relatively large clast composed of Fe-Ni, Ni-Fe, and FeS phases. The olivine clast is roughly equant in shape, 0.5 mm across, and has weak normal zoning from Fo₈₁ to Fo₇₈, averaging about Fo₈₀. This clast is distinctive both for its size, ten times greater than any of the other olivines in sample 67955,22, and its higher Fo content.

The Fe-Ni, Ni-Fe, and FeS clast has an extremely irregular shape, in contrast to the other clasts. In outline, it is approximately 0.5 by 0.2 mm. At its border occur tiny dendrites of FeS which extend into the silicate matrix, suggesting their formation was contemporaneous with the last shock; this texture is well-illustrated in Fig. 1b of Gooley *et al.*, 1973. The bulk of the grain consists mainly of Fe-Ni (Fe ≈ 90.3%; Ni ≈ 7.0%; CO ≈ 0.9%), with a small "droplet" (10 μm) at one end of Ni-Fe (Fe ≈ 42.6%; Ni ≈ 54.5%). Both phases are enveloped and separated

by FeS (Fe $\approx$ 59.2%). The Fe–Ni grain is homogeneous (no resolvable lamellae of Ni–Fe); the Ni–Fe grain is too small to detect lamellae of a second phase.

Based on a comparison of compositions of the three phases with those reported for the Ni–Fe–S phase diagram (Kullerud, 1963) and based on the textural observation that the Ni and Fe phases are separated from the silicates by an envelope of FeS, it is tentatively concluded that the predecessor of this grain has been transformed to its present shape and assemblage by a later event and re-equilibrated at a temperature below 600°C. Thus, the origin of the grain is ambiguous. It could be a meteorite fragment or, an important possibility not to be discounted at this time, a fragment of material indigenous to the lunar crust.

In one of the clasts a small hypersthene (60 μ m long) is enveloped by an irregularly-shaped grain of an Fe–Ni phase (Ni $\approx$ 6.3%) and Ni–Fe phase (Ni $\approx$ 27.0%). The Co content is between 0.4 and 0.8 wt.%. Phosphorus was not analyzed for. The closely intergrown relationship of the hypersthene and metal phases makes it hard to conceive that the hypersthene and the metal alloy phases have separate and distinct origins. Further, the hypersthene composition is similar to that in the other clasts, supporting the suggestion that the Fe–Ni and Ni–Fe phases may in part be indigenous to lunar crystal processes, as well as being in part contaminants from meteorites.

DISCUSSION

The following points may be made based on the study of the one section of sample 67955: (1) The textures and compositions of the mafic minerals in the clasts predate the last brecciation and shock-lithification events. (2) The poikilitic pyroxenes in the clasts are coarser (as measured by thickness of arms of poikocrysts) but otherwise resemble those described from other Apollo 16 samples. (3) Nevertheless, the mafic minerals in the clasts are extremely fine-grained by lunar as well as terrestrial standards. The mafic minerals (0.1 mm) are much smaller than those of typical lunar basalts (up to and larger than 1.0 mm) and even of 15415 (up to 0.3 mm). (4) The minerals are remarkably homogeneous, in contrast to those in the majority of lunar basalts. (5) The mafic silicates are notably magnesian. (6) The compositions of the pyroxenes, assuming they coexist, suggest a temperature of final chemical equilibration exceeding 900°C. (7) The spherical shape of the tiny olivine spherules included in the feldspar is noteworthy and demands an explanation. (8) The origin of the metal phases is ambiguous, but also demands an explanation.

The homogeneity of the pyroxenes suggests (1) they either grew in equilibrium with a large mafic magma reservoir and/or (2) subsequent to growth they were held at a sufficiently high temperature that homogeneous equilibrium was achieved (that is, any original inhomogeneities were removed by diffusion of ions in the minerals). The very fine grain size is inconsistent with a model of direct growth from a large mafic magma reservoir: minerals which grow in this manner tend to be much larger, even if they grow from a quenched melt. This fact has been abundantly brought out in studies of the lunar basalts. On the other hand, a

high temperature origin is supported by the distribution of Mg and Fe between the pyroxenes indicating a high temperature of *final* chemical equilibration, certainly over 900°C.

The similar composition of each mineral independent of occurrence as clast or fragment in breccia suggests (1) a common source rock for the clasts and fragments and (2) the presence of a common medium through which chemical equilibrium could be established between the minerals. This common medium, by terrestrial analogy, is either an intergranular, supercritical, low density phase, or is a silicate melt. Given the minimum estimated temperature of *final* chemical equilibration of the pyroxenes at 900°C it is reasonable to conclude the common medium was a silicate melt.

The question of whether this silicate melt phase once constituted most of the pre-brecciation rock, or whether at most only a few percent of it was molten, cannot be resolved from the limited amount of data in the present paper. The very high Mg/Fe ratio of the present rock may be an inherited characteristic from crystallization of a pre-mare lunar magma, or it may be indicative of origin of the rock as a refractory residuum resulting from removal of low-melting components by repeated impacts (Crawford and Weigand, 1973). If the pyroxene poikilitic texture has resulted from crystallization at near liquidus conditions (model of Simonds *et al.*, 1973), this melt could be the same as the pre-mare lunar magma. It could also, as proposed by Simonds *et al.* (1973), be an impact-generated melt.

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